MTH1003

UNIVERSITY OF EXETER

COLLEGE OF ENGINEERING, MATHEMATICS AND PHYSICAL SCIENCES

MATHEMATICS

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Mathematical Modelling

Module Leader: Prof John Thuburn, Prof Andrew Gilbert

Duration: 2 HOURS

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.

This is a **CLOSED NOTE** examination.

No word count specified

SECTION A

1. (a) A particle of mass $m = 1/4$ is initially at the origin with velocity $-4\mathbf{i} + 4\mathbf{j}$. It is acted upon by a time dependent force

$$\mathbf{F} = -\frac{1}{(1+t)^2}\mathbf{j} + \mathrm{e}^{-4t}\mathbf{k}.$$

Find the velocity and position of the particle as functions of time. Determine the speed of the particle in the limit $t \to \infty$. (8)

(b) The following section of Matlab code uses the bisection method to seek a root of the function $f(x)$ whose values are computed by the Matlab `function func.m`. Explain how the code must be modified in order to use the Regula Falsi method instead of the bisection method. Recall that the Regula Falsi method constructs the new value of x by fitting a straight line through the previous two data points. Explain briefly why the bisection method must converge provided $f(x)$ is continuous and the first guess brackets a root. (8)

```
% Solve func(x)=0
% First guess for interval
xl = 0; fl = func(xl);
xr = 1; fr = func(xr);
% Iterate bisection method
while xr - xl > 1e-5
    xn = 0.5*(xl + xr);
    fn = func(xn);
    if fn*fl <= 0
        % Root is in left
        % half interval
        xr = xn;
        fr = fn;
    else
        % Root is in right
        % half interval
        xl = xn;
        fl = fn;
    end
end
disp('Root lies in ')
disp(['[',num2str(xl),...
      ',',num2str(xr),']'])
```

(c) A forced oscillator is governed by

$$\ddot{x} + 9x = \cos\omega t$$

and has initial condition $x = 0$, $\dot{x} = 0$ at $t = 0$. Find the solution $x(t)$ for the two cases (i) $\omega = 2$, (ii) $\omega = 3$. (13)

(d) Write down the Forward Euler time stepping scheme for the ordinary differential equation $\dot{y} = f(y)$. Suppose the initial condition is $y(0) = y_0$. For the case $f(y) = (3\mathrm{i} - 1)y$, write down the exact solution for $y(t)$. For the same $f(y)$ compute the amplification factor for the Forward Euler time stepping scheme. Hence determine the range of step sizes for which the Forward Euler scheme is stable for this problem. (12)

(e) For the dynamical system

$$\dot{x} = 1 - x^2 - y^2, \qquad \dot{y} = xy - y^2 - x + y,$$

sketch the nullclines and, hence or otherwise, find all of the equilibrium points.

Classify the stability of the fixed point at $x = 0,\ y = 0$ for the dynamical system

$$\dot{x} = 3x - y, \qquad \dot{y} = x - 2y.$$

(9)

[**50**]

SECTION B

2. Projectiles are launched from the origin O, at speed V and angle α to the horizontal, towards a target at height h a horizontal distance $3h$ away.

   (a) Starting from the trajectory equation, which you may either quote or derive from the solutions for $x(t)$ and $z(t)$, derive the equation for the envelope of the trajectories. (8)

   (b) If a projectile is launched with speed $V = \sqrt{9gh/2}$, determine the launch angles α for which the projectile will hit the target. (5)

   (c) If a projectile is launched with $\tan\alpha = 1/2$, determine the launch speed V for which the projectile will hit the target. (4)

   (d) Determine the minimum launch speed V for which a projectile can hit the target. For this launch speed, find the slope of the trajectory $\mathrm{d}z/\mathrm{d}x$ at the point where it hits the target. (8)

   [**25**]

3. A particle of mass $m = 1$ moves along the x-axis under a conservative force given by the potential

$$V(x) = (2 - x^2)\exp(-x^2/2).$$

   (a) Find the force $F(x)$ associated with this potential.
   Calculate $V'(x)$ and $V''(x)$, and hence sketch a graph of V against x. Find the positions of the equilibrium points, and determine their stability and the frequency of small oscillations about the positions of any stable equilibria. (13)

   (b) The particle is projected from $x = 2$ with velocity $\dot{x} = U$. Write down the energy equation and find the total energy of the particle. Determine what types of behaviour are possible for different ranges of U. (8)

   (c) Discuss briefly why a forward Euler time stepping method might be unsuitable for modelling the motion of the particle in this potential. (4)

   [**25**]

4. (a) Consider an infinitely differentiable function $f(x)$. Let $x_1 = 0$, $x_2 = h$, $x_3 = 2h$, and let $f_1 = f(x_1)$, $f_2 = f(x_2)$, $f_3 = f(x_3)$.

(i) Write down the three Lagrange polynomials corresponding to the three data points. Hence write down the quadratic function $\hat{f}(x)$ that interpolates the three data points. (6)

(ii) Calculate the derivative $\hat{f}'(x)$ and hence evaluate $\hat{f}'(x_1)$. (3)

(iii) Write down Taylor series for $f(h)$ and $f(2h)$ about $x = 0$, keeping as many terms as you need. Use these to show that $\hat{f}'(x_1)$ is a second-order accurate approximation to $f'(0)$. (8)

(b) An observer $\mathcal{O}'$ moves at constant velocity u in the x-direction relative to another observer $\mathcal{O}$. In special relativity the coordinates of an event (x', t') observed by $\mathcal{O}'$ are related to the coordinates of the same event (x, t) observed by $\mathcal{O}$ by a Lorentz transformation:

$$x' = \gamma(x - ut), \qquad t' = \gamma(t - ux/c^2), \qquad \gamma(u) = (1 - u^2/c^2)^{-1/2},$$

where c is the speed of light. An alien spacecraft has length 3000 m according to its inhabitants. The spacecraft passes close to Earth at speed $4c/5$. Using the Lorentz transformation given above, determine its length as measured by an observer on Earth. (8)

[**25**]